

Test Plan & UAT Document

ReqAgent — AI Requirement Intake & Tracker

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Version: 1.0 | Date: June 2026 | Status: Final

1. Test Objectives

This document defines the test strategy, test cases, and UAT acceptance criteria for the ReqAgent system. It covers functional testing, integration testing, and user acceptance testing.

- Verify all functional requirements are implemented correctly
- Confirm AI outputs meet quality and consistency standards
- Validate dual-database architecture stores data correctly
- Ensure lifecycle management and audit trail function as specified
- Confirm traceability matrix generates accurate, complete data
- Verify analytics dashboard displays correct metrics and charts

2. Test Scope

Area	In Scope	Out of Scope
Intake Form	Field validation, submission, error handling	Browser compatibility testing
AI Processing	Story generation, MoSCoW, priority, conflict resolution	Model accuracy benchmarking
MongoDB	Insert, read, update operations	Performance under high load
Postgres	CRUD operations, joins, audit logging	Database failover testing
Dashboard	All 4 tabs, filters, status updates	Mobile responsiveness
CSV Export	Traceability matrix download	File format compatibility

3. Functional Test Cases

ID	Test Case	Expected Result	Module	Priority
TC-01	Intake Form — Required Field Validation	Validation error displayed; form not submitted	Form	High
TC-02	Intake Form — Successful Submission	Success message, AI results displayed below form	Form	High
TC-03	MongoDB — Raw Intake Storage	Document appears in MongoDB raw_intake collection with correct fields	Database	High
TC-04	AI — User Story Generation	3 user stories generated in correct As-a/I-want/So-that format	AI	High
TC-05	AI — Acceptance Criteria Format	Each story has Given/When/Then format criteria	AI	High
TC-06	AI — MoSCoW Classification	MoSCoW classification assigned and displayed	AI	High
TC-07	AI — Priority Scoring	Priority shows 'Low' (role-weighted)	AI	High
TC-08	AI — Conflict Detection	Conflict flagged with description	AI	High
TC-09	Postgres — Requirement Storage	Record appears in requirements table with correct data	Database	High
TC-10	Postgres — User Story Storage	3 rows appear in user_stories table linked to requirements	Database	High
TC-11	Tracker — Display All Requirements	All submitted requirements listed with status badges	Tracker	High
TC-12	Tracker — Filter by Status	Only Approved requirements shown	Tracker	Medium
TC-13	Tracker — Status Update	Status updated in Postgres; audit log entry created	Tracker	High
TC-14	Tracker — Audit Trail	Both status changes shown in audit trail with timestamps	Tracker	High
TC-15	Traceability — Matrix Display	All requirements with user stories displayed in table	Traceability	High
TC-16	Traceability — Filter by MoSCoW	Only Must Have requirements shown in matrix	Traceability	Medium
TC-17	Traceability — CSV Export	CSV file downloaded with all matrix columns	Traceability	Medium
TC-18	Analytics — KPI Display	Total, Approved, In Dev, Done, High Priority, Must Have counts shown	Analytics	High
TC-19	Analytics — Charts Rendered	Status donut, MoSCoW pie, department bar, priority heatmap, role chart rendered	Analytics	Medium
TC-20	Error Handling — API Failure	User sees error message; raw intake still saved to MongoDB	Database	High

4. UAT Acceptance Criteria

Criterion	Measure	Pass Threshold
Form submission success rate	% of submissions that complete without error	100%
AI user story quality	% of stories in correct format (As-a/I-want/So-that)	95%+
AI MoSCoW accuracy	% of classifications matching expected output	90%+
Priority role-weighting accuracy	% of priorities matching role-weight mapping	100%

Criterion	Measure	Pass Threshold
MongoDB storage success rate	% of submissions stored correctly	100%
Postgres storage success rate	% of requirements + stories stored correctly	100%
Audit trail completeness	% of status changes logged	100%
Traceability matrix accuracy	% of rows matching requirements + stories in DB	100%
CSV export functionality	CSV downloads successfully with all columns	100%
Analytics data accuracy	KPI counts match actual database counts	100%
AI processing time	Time from submission to results display	< 10 seconds
Page load time	Time for any tab to load	< 3 seconds

5. UAT Sign-Off Checklist

Item	Verified By	Status	Date
All 20 functional test cases passed	BA Lead		
All UAT acceptance criteria met	Product Owner		
MongoDB data verified in Atlas console	Database Admin		
Postgres data verified via direct query	Database Admin		
CSV export reviewed for accuracy	Business Analyst		
Analytics charts verified against DB counts	Business Analyst		
Error handling scenarios tested	QA Lead		
Performance benchmarks met	QA Lead		
Security review completed (no exposed credentials)	Security Lead		
Final stakeholder demo completed	Project Manager		